

CIS — Z Bank Online v1.0

Component Integration Specification

Field	Value
Document ID	CIS-Z-BANK-ONLINE-v1
Version	1.0
Status	Draft
Platform	OneX Bridge / Online Banking
Customer brand	Z Bank
Internal service	onex-production-platform / onex-ledger-middleware
Peer brand	Nova Bank Online — see CIS-Nova-Bank-Online-v1.md (M0/M1/NSB; leave unchanged)

1. Purpose and scope

This CIS defines how **Z Bank** uses the OneX production platform with an **internal M1–M4 fiat liquidity architecture**. These layers are Z Bank operational fund classes for wallets, safeguarded settlement, treasury, and wholesale/tokenized liquidity. They are **not** public macroeconomic money aggregates, and they are **not** account types advertised by Wise, Revolut, or other EMIs.

In scope

- Z Bank ledger seed accounts (M0 reserves + M1–M4 operational layers)
- Payment gateway framework `zbank` with settlement into Z Bank accounts
- Fund-class aliases shared with the ledger middleware
- REST API contract on the OneX bridge (:9338) when Z Bank env is active
- Corporate officer signatory auth (PIN + signature) — see CIS-Z-Bank-DSSBOAT-Officer-v1.md

Out of scope

- Changing Nova Bank seeds, CIS, or production defaults
- Full KYC/AML vendor selection
- Live SWIFT/SEPA scheme execution beyond existing rails wiring

2. System identifiers

Identifier	Value
Bank display name	Z Bank
Payment gateway framework	<code>zbank</code>
Bank ID (routing / seed bank)	<code>zbank</code>
Bridge listen port	9338
Wallet UI path	<code>/wallet/#onlinebank</code>
Payments UI path	<code>/payments/</code>

3. Operational fund classes (M1–M4)

Class	Label	Z Bank meaning	Seed account examples
m0	M0 — base reserves	Cash vault / central-style reserves (optional support layer)	zbank-usd-cash
m1	M1 — spendable wallet	Customer immediately spendable fiat (current / card balance)	zbank-usd-checking , zbank-eur-checking
m2	M2 — safeguarded settlement	Reserved / safeguarded balances awaiting payout or release	zbank-usd-safeguarded
m3	M3 — treasury liquidity	Treasury / liquidity pool for interbank or institutional ops	zbank-usd-treasury
m4	M4 — wholesale / tokenized	Cross-border settlement, tokenized fiat, wholesale liquidity	zbank-usd-wholesale , zbank-gbp-wholesale

3.1 Alias compatibility

Ledger `NormalizeFundClass` keeps prior synonyms (`savings` → m2, `institutional` → m3, `money_market` → m4) and adds Z Bank operational aliases:

Target	Aliases (non-exhaustive)
m2	safeguarded , settlement_reserve , reserved (+ existing savings)
m3	treasury , liquidity_pool (+ existing wholesale , institutional)
m4	cross_border , wholesale_liquidity , tokenized_fiat (+ existing money_market , repo)

3.2 Layer flow

[Architecture diagram — see markdown source]

```

flowchart LR
    subgraph zbank [Z Bank layers]
        M1[M1 spendable wallet]
        M2[M2 safeguarded settlement]
        M3[M3 treasury liquidity]
        M4[M4 wholesale tokenized]
    end
    M1 -->|card or payout reserve| M2
    M2 -->|ops funding| M3
    M3 -->|batch settle or mint| M4
    M4 -->|fiat middleware| ONEXUSD[ONEXUSD mint pool]

```

Fiat settlement middleware continues to aggregate pools `pool:m1 ... pool:m4` and `pool:mint` (see ledger middleware). M0 source balances route into the M1 pool before aggregate settlement.

4. Data models

4.1 Online / bank ledger account

Field	Type	Description
id	string	Account ID (e.g. zbank-usd-checking)
name	string	Display name
iban	string	IBAN or Z Bank-format account number
currency	string	ISO 4217
balance	string	Decimal balance
fundClass / moneySupply	string	m0 , m1 , m2 , m3 , m4
bank	string	zbank

Seed data: configs/bank-ledger.zbank.example.json

4.2 Payment gateway settlement

Default gateway file: configs/payment-gateway.zbank.example.json

Destination id	Internal accountId	Layer
zbank-usd-main	zbank-usd-checking	M1
zbank-usd-safeguarded	zbank-usd-safeguarded	M2
zbank-usd-treasury	zbank-usd-treasury	M3

Card **pay** pages settle to M1; **collect** pages settle to M2 (safeguarded). Do not settle Z Bank payments into Nova account IDs.

5. Environment

Template: deploy/env.zbank.example

```
ONEX_LEDGER_MODE=production
ONEX_ONLINE_BANK=1
ONEX_BANK_LEDGER_FILE=configs/bank-ledger.zbank.production.json
ONEX_PAYMENT_GATEWAY=1
ONEX_PAYMENT_GATEWAY_FILE=configs/payment-gateway.zbank.production.json
ONEX_PAYMENT_GATEWAY_FRAMEWORK=zbank
ONEX_PAYMENT_GATEWAY_PROVIDER=stripe
ONEX_ZBANK_OFFICERS_FILE=configs/zbank-officers.dssboat.example.json
ONEX_ZBANK_OFFICER_PIN=<4-8-digit-pin>
ONEX_ZBANK_OFFICER_SIGNATURE=<min-8-char-signature>
ONEX_API_KEY=<long-random-secret>
```

Full production template: deploy/env.zbank.production.example | Environment | Host | Notes | |-----|-----|-----| |
 Local dev | 127.0.0.1:9338 | Use zbank ledger + PG files above | | Production | Operator-selected domain | Set
 FRAMEWORK=zbank ; keep Nova production defaults on separate hosts |

6. API contract (shared bridge)

Base URL: https://<HOST>/bridge (or http://127.0.0.1:9338/bridge locally)

When Z Bank env is loaded, the same online-bank and ledger routes apply:

Method	Endpoint	Purpose
GET	/bridge/bank/status	Bank health
GET	/bridge/bank/accounts	List accounts (expect M1–M4 Z Bank IDs)
GET	/bridge/ledger/status	Middleware status including fund class labels
GET	/bridge/bank/officer/status	DSSBOaT / corporate officer store
POST	/bridge/bank/officer/verify	Verify officer PIN + signature
POST	/bridge/bank/officer/transfer	Officer-authorized transfer (PIN + signature)
POST	/bridge/payments/*	Card flows via Z Bank gateway
POST	/bridge/ledger/middleware/fiat-settle	Batch M1–M4 → mint aggregation

Corporate officer CIS: CIS-Z-Bank-DSSBOaT-Officer-v1.md

Authentication: X-API-Key: <ONEX_API_KEY> on mutating endpoints where enforced.

7. Contrast with Nova Bank

Topic	Nova Bank	Z Bank
CIS	CIS-Nova-Bank-Online-v1.md	This document
Ledger seed	bank-ledger.nova.example.json	bank-ledger.zbank.example.json
Fund model	M0 / M1 / NSB	M0 + M1–M4 operational layers
PG framework	nova	zbank
Settlement IDs	nova-*	zbank-* only

Nova and Z Bank share the OneX bridge code path; switch brand via env and config files, not by mixing account IDs across brands.

8. Verify after deploy

```
curl -s https://HOST/bridge/bank/status | jq .
curl -s https://HOST/bridge/bank/accounts | jq .
curl -s https://HOST/bridge/ledger/status | jq '.fundClasses, .fundClassLabels'
curl -s https://HOST/payments/ | head
```

Expect ledger status to list m0 – m4 (and related classes) and bank accounts to include zbank-usd-checking, zbank-usd-safeguarded, zbank-usd-treasury, and wholesale (M4) IDs.